ASU Micro Qual Module 2
General Equilibrium, Excess Demand, and Welfare Theorems

PhD Qual Lecture Notes Builder Packet

Compiled for exam-facing study

Contents

1 Orientation

This packet is a self-contained, derivation-heavy module for the ASU Microeconomic Theory qualifying exam. The target is not merely to remember definitions, but to be able to write a complete exam answer under time pressure. The module emphasizes five skills:

1. defining a general equilibrium object cleanly;
2. deriving individual and aggregate excess demand;
3. using Walras' Law to reduce market clearing and detect mistakes;
4. computing Pareto optima and Walrasian equilibria in Edgeworth-box and Robinson Crusoe economies; and
5. proving the First and Second Welfare Theorems with exact assumptions.

Exam-facing rule

For almost every problem in this module, use the four-layer answer: **Object** → **Feasibility** → **Optimality/Equilibrium** → **Verification**. Do not start with FOCs until the object and feasibility constraints are explicit.

Expected study use. One deep 5–6 hour block for definitions, existence, and welfare theorems; one 4 hour problem-solving block for Edgeworth boxes, excess demand, and Robinson Crusoe computations; then one 2 hour old-qual drill.

2 Source Inventory

The source hierarchy used here is: past ASU Micro qualifying exams determine exam relevance; current general-equilibrium problem sets determine current-course emphasis; professor lecture notes provide notation and proof framing; standard MWG/Kreps theory fills in gaps only where needed.

Source	Type	Module content extracted	Extraction notes
MICRO_MANIFEST.md	Manifest	Locked Module 2 scope; source hierarchy; notation for budget sets, Walrasian demand, indirect utility, and excess demand; required derivations and output sections.	Fully readable.
SKILL.md	Builder skill	Required workflow: source inventory, pattern analysis, blueprint, standard packet, exercise distribution, and quality audit.	Fully readable.

Source	Type	Module content extracted	Extraction notes
GitHub: Microeconomics/General Equilibrium/dge01-intro-defns.tex	Lecture notes	Arrow-Debreu economy, allocation, feasibility, Pareto optimality, budget sets, demand sets, Walrasian equilibrium with transfers.	Raw TeX readable; some source file formatting compressed.
GitHub: dge02-graphs.tex	Lecture notes	Edgeworth-box construction, feasibility as box points, budget line geometry, graphical PO, boundary and kink warnings, Robinson Crusoe graph framing.	Raw TeX readable; original graphics not imported, so packet uses new TikZ and verbal graph protocols.
GitHub: dge03-fwt.tex	Lecture notes	First Welfare Theorem proof: preference maximization implies price inequalities; LNS converts weak preference to weak price inequality; aggregate contradiction using profit maximization.	Raw TeX readable.
GitHub: dge040-swt.tex	Lecture notes	Separating-hyperplane proof of Weak Second Welfare Theorem; preferred sets; quasi-equilibrium; transfer construction.	Raw TeX readable.
GitHub: dge050-fpt-correspondences.tex	Lecture notes	Correspondence definitions, fixed point definitions, Kakutani conditions, UHC/closed graph tools.	Raw TeX readable.
GitHub: dge060-exwe.tex	Lecture notes	Aggregate excess demand on simplex interior, Walras' Law, boundedness below, boundary condition, DGKN existence proof outline, missing-market-clearing trick.	Raw TeX readable.
GitHub: dge092-core.tex	Lecture notes	Core definition, blocking coalitions, core implies PO, and WE allocations in core under monotonicity.	Raw TeX readable; used as optional extension.
GitHub: dge100-unc-sec.tex	Lecture notes	State-contingent commodities, Arrow-Debreu contingent claims, complete markets under uncertainty, ex ante PO interpretation.	Raw TeX readable.

Source	Type	Module content extracted	Extraction notes
GE Problem Set 1, ECN 714	Problem set	Nonstandard consumption sets; Edgeworth boxes outside the positive orthant; price-zero equilibria; integer indivisibilities; private ownership with labor/food; compute PO and WE graphically.	PDF text parsed with math loss; problem wording readable.
GE Problem Set 2, ECN 714	Problem set	Production economy calculations; 3-good Cobb-Douglas exchange; PO via planner problem; converse separating-hyperplane theorem; Edgeworth examples with boundary solutions.	PDF text parsed with math loss; equations reconstructed where needed.
GE Problem Set 3, ECN 714	Problem set	Direct SWT and existence in one-consumer production economy; UHC/closed graph exercises; boundary condition intuition; examples showing failure of continuity, Walras' Law, boundedness below, boundary condition.	PDF text parsed with math loss; main tasks readable.
GE Problem Set 4, ECN 714	Problem set	Boundary-condition proof for aggregate excess demand; Gorman aggregation; core allocations; demand-set proof using infimum of affordable strictly preferred bundles.	PDF text parsed with math loss.
GE Problem Set 5, ECN 714	Problem set	Equilibrium correspondence UHC and limit equilibrium; identical-consumer WE examples with nonconvex preferences.	PDF text parsed with math loss.
ASU August 2024 Q1	Past qual	Compute aggregate excess demand for two-agent square-root exchange economy; identify WE price; state existence conditions; provide no-WE example.	PDF parsed; high relevance.

Source	Type	Module content extracted	Extraction notes
ASU June 2020 Q3–Q4	Past qual	Same square-root exchange computation; no-WE example; abstract economy with auctioneer and relation to competitive equilibrium.	PDF parsed; high relevance.
ASU June 2021 Q1	Past qual	Test an excess demand function against continuity, homogeneity, Walras' Law, boundedness below, boundary condition; prove no WE.	PDF parsed; high relevance.
ASU August 2018 Q3–Q4	Past qual	Closed-graph proof for equilibrium price correspondence; state-contingent Edgeworth box and risk-sharing with aggregate no-risk.	PDF parsed; high relevance.
ASU August 2015 Q1	Past qual	Edgeworth-box WE, Pareto optimality proof, and direct construction of WE with transfers for a proposed PO allocation.	Scan visible; OCR limited.
ASU August 2017 Q1–Q2	Past qual	Walras' Law missing-market-clearing proof in private ownership economy; FWT-style demand inequality.	Scan visible; OCR limited.
ASU June 2022 Q1	Past qual	Robinson Crusoe economy with production set $Y = \mathbb{R}^L_-$; WE existence/uniqueness issues; SFWT assumptions versus conclusion.	PDF parsed.
ASU June 2023 Q3	Past qual	Robinson Crusoe economy with linear technology and quasilinear-sqrt utility; derive supply, demand, all CE; representative-firm discussion.	PDF parsed.
ASU August 2023 Q1	Past qual	State-contingent Edgeworth-box PO, risk sharing, boundary counterexample.	PDF parsed.
ASU June 2016 Q1	Past qual	Pure exchange with a non-tradable good; constrained versus unconstrained PO.	Scan visible; OCR limited.

Not used heavily. GitHub notes on uniqueness, genericity, and stability were inventoried from the folder listing but are lower probability for this module's requested target. They are mentioned only in the final checklist as optional extensions.

3 Pattern Analysis and Detailed Blueprint

3.1 Core exam archetypes

Archetype	Signature wording	Reusable move	Trap
Compute excess demand and WE	"Find the aggregate excess demand function"; "Use $z(p)$ to find WE price."	Solve each consumer problem, impose budget exhaustion, aggregate $x_i - \omega_i$, normalize one price or use Δ.	Forgetting wealth is $p \cdot \omega_i$, not a fixed number.
Test existence conditions	"Give conditions that guarantee WE exists"; "Does z satisfy continuity, Walras' Law, boundedness below, boundary?"	DGKN checklist: continuity, Walras' Law, boundedness below, boundary condition; then $z(p^*) = 0$.	Using only $L - 1$ equations without Walras' Law and positive prices.
No-WE example	"Give an economy for which no WE exists" or "Does this economy have a WE?"	Try relative-price cases, use demand behavior, show every case violates some market.	Saying "nonconvex preferences" alone proves nonexistence. It does not.
Walras' Law trick	"Assume first $L-1$ markets clear; prove the last clears."	Sum budget equalities and firm zero/profit identities to get $p \cdot z = 0$; then $p_L > 0$ forces $z_L = 0$.	Forgetting transfers and firm profits in private ownership economies.
Closed graph / UHC	"Define closed graph"; "Prove equilibrium correspondence has closed graph."	Sequential definition: $\omega^n \to \omega$, $p^n \to p$, $F(p^n, \omega^n) = 0$, continuity gives $F(p, \omega) = 0$.	Trying to prove compact-valued or UHC when only closed graph is asked.
Edgeworth PO	"Calculate Pareto set"; "illustrate in an Edgeworth box."	Planner problem: maximize u_1 subject to $u_2 \geq \bar{u}_2$ and feasibility; interior FOC $MRS_1 = MRS_2$; check boundaries.	Equating MRS at a boundary, kink, Leontief corner, or linear preference case.

Archetype	Signature wording	Reusable move	Trap
Risk-sharing PO	"One good, two states"; "aggregate endowment same/different across states."	Treat states as goods; common beliefs and concavity give equalized weighted marginal utilities.	Confusing ex ante contingent-claim PO with ex post allocation after state realization.
FWT proof	"Prove every WE is PO"; "what role does LNS play?"	Better bundles must be more expensive; weakly better bundles cannot be cheaper; aggregate strict inequality contradicts feasibility plus profit maximization.	Using convexity. FWT does not need convexity.
SWT/SFWT proof	"Support a PO as WE with transfers"; "do assumptions hold?"	Separate preferred aggregate set from feasible production set; construct prices; construct transfers so each x_i^* is just affordable.	Claiming a true WE rather than a quasi-equilibrium without the extra cheaper-bundle/slack condition.
Robinson Crusoe	"Define CE for one consumer, one firm"; "derive supply and demand."	Firm profit max, consumer demand with profit income, feasibility $x = \omega + y$.	Forgetting the consumer owns the firm, so profits enter wealth.

3.2 Module objective

By the end of this packet, the student should be able to take any ASU GE question and, within five minutes, classify it into one of the archetypes above, write the correct definitions, and begin the correct derivation without searching memory for the theorem statement.

3.3 Prerequisites

The packet assumes comfort with consumer optimization, KKT conditions, convexity, separating hyperplanes, and basic correspondence language. Every GE-specific use of these tools is rederived below.

3.4 Canonical models covered

1. finite Arrow-Debreu exchange and private-ownership economies;
2. two-good two-agent exchange economy and Edgeworth box;
3. aggregate excess demand and normalized price simplex;
4. DGKN existence theorem for excess demand;
5. state-contingent exchange economy under complete markets;

6. Robinson Crusoe production economy; and
7. core as a short optional check on voluntary trade and PO.

3.5 Closed-form catalog

The packet derives these high-yield closed forms.

1. Square-root demand: if $u(x_1, x_2) = \sqrt{x_1} + \sqrt{x_2}$ and wealth is w, then
$$x_1(p, w) = \frac{wp_2}{p_1(p_1 + p_2)}, \qquad x_2(p, w) = \frac{wp_1}{p_2(p_1 + p_2)}.$$
2. ASU 2020/2024 aggregate excess demand for endowments $(2, 0), (0, 2)$:
$$z(p) = \left(2\frac{p_2}{p_1} - 2,\ 2\frac{p_1}{p_2} - 2\right), \qquad p_1 = p_2 \text{ in every WE.}$$
3. Two-good Cobb-Douglas demand: if $u_i(x) = \alpha_i \log x_1 + (1 - \alpha_i) \log x_2$, then
$$x_{i1} = \alpha_i \frac{p \cdot \omega_i}{p_1}, \qquad x_{i2} = (1 - \alpha_i)\frac{p \cdot \omega_i}{p_2}.$$
4. State-contingent risk-sharing FOC at an interior PO:
$$\frac{u_i'(x_1^i)}{u_i'(x_2^i)} = \frac{\nu_1/\pi_1}{\nu_2/\pi_2} \quad \text{for all } i,$$
where ν_s is the resource multiplier in state s.
5. Robinson Crusoe linear technology $q = Az$, utility $2\sqrt{x_1} + x_2$, price of input good normalized to 1:
$$p^* = \frac{1}{A}, \qquad z^* = \min\{A, \omega\}, \qquad q^* = Az^*.$$

4 Module Map

Subtopic	**Why it matters**	**Prototype**	**Core technique**	**Common trap**
Definitions of WE and PO	Frequent first part; sets notation for all proofs.	2015 Q1, 2020 Q4, 2022 Q1, 2023 Q3.	Object–feasibility–optimality–market clearing.	Omitting firm ownership/profits.
Excess demand	Most common computational GE problem.	2020 Q3, 2024 Q1.	Individual demand $\to$ aggregate $z(p)$.	Wealth is endogenous.
Walras' Law	Lets one market be omitted; verifies algebra.	2017 Q1, 2020 Q4.	Sum budget equalities.	Applying with nonpositive price of missing good.

Subtopic	Why it matters	Prototype	Core technique	Common trap
Existence conditions	ASU asks theorem hypotheses directly.	2021 Q1, 2024 Q1.	DGKN: continuity, WL, bounded below, boundary.	Boundary condition misread as "one component bounded."
Closed graph	Short proof question.	2018 Q3, PS5.	Sequential continuity proof.	Proving UHC without compactness.
Edgeworth box	Required graphical tool.	PS1, PS2, 2015 Q1.	Contract curve and budget line.	MRS equality at boundaries.
PO computation	Planner problems recur in exchange, risk, and production.	PS2, 2023 Aug Q1.	KKT with resource constraints.	Ignoring nonnegativity.
FWT	High-yield proof template.	2017 Q2, lectures.	Price inequality contradiction.	Using convexity.
SWT/SFWT	Assumption-check favorite.	2015 Q1, 2022 Q1, PS3.	Separate preferred aggregate set from feasible set.	Forgetting transfers.
Robinson Crusoe	Bridges GE and production.	2014 J Q1, 2022 J Q1, 2023 J Q3.	Firm supply + consumer demand + feasibility.	Ignoring unbounded profit cases.
State-contingent GE	Cross-cutting uncertainty skill.	2018 Q4, 2023 A Q1, 2024 A Q4.	Treat states as goods.	Ex ante vs ex post confusion.
Core	Optional but recurring in PS and some exams.	PS4, 2019 Q2 analogy.	Blocking coalition definitions.	Calling every PO core.

5 Learning Objectives

After studying this module, you should be able to:

1. state the definition of an exchange economy, private-ownership economy, allocation, PO, WE, WE with transfers, and quasi-equilibrium;
2. derive an individual demand correspondence and aggregate excess demand function, including endowment wealth $p \cdot \omega_i$;
3. prove Walras' Law from budget exhaustion and use it to infer missing market clearing;
4. verify or refute continuity, homogeneity, Walras' Law, boundedness below, and boundary condition for a proposed z;
5. compute the Pareto set in an Edgeworth box with smooth, linear, Leontief, quasilinear, and

boundary cases;

6. draw the graph that corresponds to a budget line, allocation, PO, WE, transfer-supported WE, or no-WE case;
7. prove FWT by contradiction with the two price-inequality implications;
8. prove the support step in SWT using separating hyperplanes and then construct transfers;
9. solve a Robinson Crusoe economy by combining firm supply, consumer demand, and feasibility; and
10. write concise answer sketches for ASU-style questions without overclaiming assumptions.

6 Core Definitions

6.1 Arrow-Debreu and pure exchange economies

Definition 1 (Arrow-Debreu economy). *A finite private-ownership Arrow-Debreu economy is*

$$\mathcal{E} = \left(L, \{Y_j\}_{j=1}^J, \{X_i, \succeq_i, \omega_i, \{\theta_{ij}\}_{j=1}^J\}_{i=1}^I\right),$$

where L is the number of commodities, $Y_j \subseteq \mathbb{R}^L$ is firm j's production set, $X_i \subseteq \mathbb{R}^L$ is consumer i's consumption set, $\succeq_i$ is consumer i's preference relation, $\omega_i \in \mathbb{R}^L$ is consumer i's initial endowment, and $\theta_{ij} \in [0,1]$ is consumer i's share of firm j with $\sum_i \theta_{ij} = 1$.

Definition 2 (Pure exchange economy). *A pure exchange economy is the special case with no firms:*

$$\mathcal{E} = \left(L, \{X_i, \succeq_i, \omega_i\}_{i=1}^I\right).$$

Usually in exam problems $X_i = \mathbb{R}^L_+$ and prices are $p \in \mathbb{R}^L_{++}$ or normalized to Δ.

Definition 3 (Feasible allocation). *For a private-ownership economy, an allocation is $(x, y) = ((x_i)_{i=1}^I, (y_j)_{j=1}^J)$ with $x_i \in X_i$, $y_j \in Y_j$, and*

$$\sum_{i=1}^I x_i = \sum_{i=1}^I \omega_i + \sum_{j=1}^J y_j.$$

In pure exchange, feasibility is simply $\sum_i x_i = \sum_i \omega_i$.

Definition 4 (Pareto optimality). *A feasible allocation (x, y) is strongly Pareto optimal if there is no feasible allocation (x', y') such that $x_i' \succeq_i x_i$ for all i and $x_k' \succ_k x_k$ for at least one k. It is weakly Pareto optimal if there is no feasible allocation (x', y') such that $x_i' \succ_i x_i$ for all i.*

Definition 5 (Budget and demand). *Given prices p, production plans y, and transfers T_i with $\sum_i T_i = 0$, consumer i's budget set is*

$$B_i(p, y, T_i) = \left\{ x_i \in X_i : p \cdot x_i \leq p \cdot \omega_i + T_i + \sum_{j=1}^J \theta_{ij} p \cdot y_j \right\}.$$

The demand set is

$$D_i(p, y, T_i) = \left\{ x_i \in B_i(p, y, T_i) : x_i \succeq_i x_i' \text{ for all } x_i' \in B_i(p, y, T_i) \right\}.$$

In pure exchange without transfers, write $B_i(p) = \{x_i \in X_i : p \cdot x_i \leq p \cdot \omega_i\}$ and $D_i(p) = D_i(p, \omega_i)$.

Definition 6 (Walrasian equilibrium). *A Walrasian equilibrium with transfers is* (x^*, y^*, p^*, T^*) *such that:*

1. $\sum_i T_i^* = 0$;
2. *every firm maximizes profit:* $p^* \cdot y_j^* \geq p^* \cdot y_j$ *for every* $y_j \in Y_j$ *and every* j;
3. *every consumer maximizes preferences:* $x_i^* \in D_i(p^*, y^*, T_i^*)$ *for every* i;
4. *markets clear:* $\sum_i x_i^* = \sum_i \omega_i + \sum_j y_j^*$.

If $T_i^* = 0$ *for all* i*, this is simply a Walrasian equilibrium. In a pure exchange economy, the firm condition disappears.*

Definition 7 (Price normalization). *If demand is homogeneous of degree zero in prices and wealth, multiplying all prices by* $\lambda > 0$ *does not change real budget sets. Hence prices can be normalized, for example by setting* $p_L = 1$ *or by imposing*

$$\Delta = \left\{ p \in \mathbb{R}_+^L : \sum_{\ell=1}^{L} p_\ell = 1 \right\}, \qquad \Delta^\circ = \{p \in \Delta : p \gg 0\}.$$

Qualifier trap

A price normalization is not an extra equilibrium condition. If the exam says normalize $p_L = 1$, do not also impose $\sum_\ell p_\ell = 1$ unless the problem explicitly asks for both.

6.2 Aggregate excess demand

Definition 8 (Aggregate excess demand). *In a pure exchange economy, if each consumer has demand* $x_i(p) \in D_i(p, p \cdot \omega_i)$*, aggregate excess demand is*

$$z(p) = \sum_{i=1}^{I} (x_i(p) - \omega_i) .$$

If demand is a correspondence, then $z(p)$ *is a correspondence. Many ASU questions assume a single-valued* z *for simplicity.*

Lemma 1 (Walras' Law). *If every consumer exhausts the budget at* $p \gg 0$*, then*

$$p \cdot z(p) = 0.$$

Proof. For each i, local nonsatiation or strong monotonicity implies $p \cdot x_i(p) = p \cdot \omega_i$. Summing across consumers,

$$\sum_i p \cdot x_i(p) = \sum_i p \cdot \omega_i.$$

Rearrange:

$$p \cdot \sum_i (x_i(p) - \omega_i) = p \cdot z(p) = 0.$$

□

Lemma 2 (Missing market clearing). *Suppose* $p \gg 0$ *and* $p \cdot z(p) = 0$*. If* $z_\ell(p) = 0$ *for every* $\ell \neq k$*, then* $z_k(p) = 0$.

Proof. Walras' Law gives

$$0 = p \cdot z(p) = p_k z_k(p) + \sum_{\ell \neq k} p_\ell z_\ell(p) = p_k z_k(p).$$

Since $p_k > 0$, $z_k(p) = 0$. $\square$

Exam-facing rule

When a problem says "assume the first $L-1$ markets clear," the expected proof is usually one line: use Walras' Law and $p_L > 0$. In production economies, derive Walras' Law from consumer budget exhaustion *including profit income and transfers.*

6.3 The four-layer answer template

Reusable template

General equilibrium answer template.

1. **Object.** State whether the object is a price, allocation, excess demand function, PO allocation, WE, WE with transfers, or correspondence.
2. **Feasibility.** Write $x_i \in X_i$, $y_j \in Y_j$, and $\sum_i x_i = \sum_i \omega_i + \sum_j y_j$.
3. **Optimality/equilibrium.** Write consumer optimality, firm profit maximization, KKT/-FOC conditions, or support inequalities.
4. **Verification.** Check budget exhaustion, market clearing, nonnegativity/boundaries, Walras' Law, and theorem assumptions.

7 Graphical Toolkit

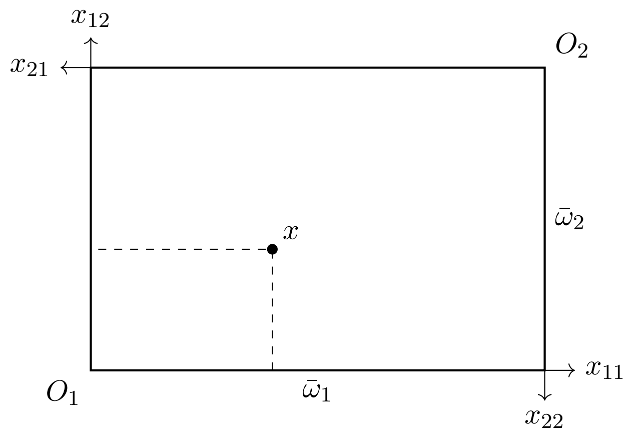
7.1 Edgeworth box protocol

For a two-agent two-good pure exchange economy, let total endowment be

$$\bar{\omega} = (\bar{\omega}_1, \bar{\omega}_2) = \omega_1 + \omega_2.$$

Draw a rectangle of width $\bar{\omega}_1$ and height $\bar{\omega}_2$. Consumer 1's origin is the southwest corner; consumer 2's origin is the northeast corner. A point in the box is an allocation: if consumer 1 gets $x_1 = (a, b)$, consumer 2 gets

$$x_2 = (\bar{\omega}_1 - a, \bar{\omega}_2 - b).$$



Budget line in the box. The budget line through endowment ω_1 is

$$p \cdot x_1 = p \cdot \omega_1.$$

Its slope in consumer 1 coordinates is $-p_1/p_2$. The same line is consumer 2's budget line because

$$p \cdot x_2 = p \cdot (\bar{\omega} - x_1) = p \cdot \omega_2 \quad \Longleftrightarrow \quad p \cdot x_1 = p \cdot \omega_1.$$

7.2 How to draw PO, WE, and transfers

- **PO.** Draw indifference curves for both consumers. Interior smooth PO points are tangencies. Boundary PO points may have no tangency.
- **WE.** Draw the budget line through the endowment. A WE allocation is a feasible point where each consumer's chosen bundle lies on this common budget line and maximizes on the relevant side of the line.
- **WE with transfers.** The supporting price line need not pass through the original endowment. Transfers shift wealth so that the line passes through the target allocation and each consumer optimizes there.

Qualifier trap

In an Edgeworth box, the budget line may extend outside the box. The feasible allocation must be inside the box, but a consumer's budget set in commodity space is not literally the box.

7.3 Interior versus boundary PO

For smooth strictly monotone preferences, an interior PO solves a planner problem such as

$$\max_{x_1,x_2} u_1(x_1) \quad \text{s.t.} \quad u_2(x_2) \geq \bar{u}_2, \qquad x_1 + x_2 = \bar{\omega}, \qquad x_i \geq 0.$$

If $x_i \gg 0$ and the utility constraint binds, the Lagrangian is

$$\mathcal{L} = u_1(x_1) + \lambda\big[u_2(\bar{\omega} - x_1) - \bar{u}_2\big].$$

The FOCs imply

$$\frac{\partial u_1/\partial x_1}{\partial u_1/\partial x_2} = \frac{\partial u_2/\partial x_1}{\partial u_2/\partial x_2},$$

which is $MRS_1 = MRS_2$.

Mathematical intuition

At an interior PO, the planner cannot transfer a tiny amount of good 1 for good 2 between agents and make one better off without hurting the other. Equal MRS says both agents have the same local exchange rate between goods.

Qualifier trap

MRS equality is not a definition of Pareto optimality. It is a first-order characterization under smoothness and interiority. Linear preferences, Leontief preferences, kinks, and boundary allocations require inequalities or direct dominance arguments.

8 Canonical Model I: Computing Excess Demand

8.1 General procedure

Given u_i, ω_i, and $p \gg 0$:

1. Compute wealth $w_i(p) = p \cdot \omega_i$.
2. Solve
$$x_i(p) \in \arg\max_{x_i \in X_i} \{u_i(x_i) : p \cdot x_i \leq w_i(p)\}.$$
3. If preferences are LNS, impose $p \cdot x_i(p) = w_i(p)$.
4. Aggregate:
$$z(p) = \sum_i x_i(p) - \sum_i \omega_i.$$
5. Verify Walras' Law: $p \cdot z(p) = 0$. If it fails, your algebra is wrong or budget exhaustion failed.

8.2 ASU square-root exchange economy

Two agents have

$$u_i(x_{i1}, x_{i2}) = \sqrt{x_{i1}} + \sqrt{x_{i2}}, \qquad \omega_1 = (2, 0), \quad \omega_2 = (0, 2).$$

Let $p = (p_1, p_2) \gg 0$.

Step 1: individual demand. For a consumer with wealth w, solve

$$\max_{x_1, x_2 \geq 0} \sqrt{x_1} + \sqrt{x_2} \quad \text{s.t.} \quad p_1 x_1 + p_2 x_2 \leq w.$$

The solution is interior if $w > 0$. The FOCs are

$$\frac{1}{2\sqrt{x_1}} = \lambda p_1, \qquad \frac{1}{2\sqrt{x_2}} = \lambda p_2.$$

Divide them:

$$\frac{\sqrt{x_2}}{\sqrt{x_1}} = \frac{p_1}{p_2} \quad \Longrightarrow \quad x_2 = \left(\frac{p_1}{p_2}\right)^2 x_1.$$

Budget exhaustion gives

$$
\begin{aligned}
p_1 x_1 + p_2 \left(\frac{p_1}{p_2}\right)^2 x_1 &= w, \\
x_1 \left(p_1 + \frac{p_1^2}{p_2}\right) &= w, \\
x_1(p, w) &= \frac{w p_2}{p_1(p_1 + p_2)}, \\
x_2(p, w) &= \frac{w p_1}{p_2(p_1 + p_2)}.
\end{aligned}
$$

Step 2: endowment wealth. Agent 1 has $w_1 = 2p_1$, so

$$
x_1^1(p) = \frac{2p_2}{p_1 + p_2}, \qquad x_2^1(p) = \frac{2p_1^2}{p_2(p_1 + p_2)}.
$$

Agent 2 has $w_2 = 2p_2$, so

$$
x_1^2(p) = \frac{2p_2^2}{p_1(p_1 + p_2)}, \qquad x_2^2(p) = \frac{2p_1}{p_1 + p_2}.
$$

Step 3: aggregate and subtract endowments.

$$
\begin{aligned}
X_1(p) &= \frac{2p_2}{p_1 + p_2} + \frac{2p_2^2}{p_1(p_1 + p_2)} = \frac{2p_2}{p_1}, \\
X_2(p) &= \frac{2p_1^2}{p_2(p_1 + p_2)} + \frac{2p_1}{p_1 + p_2} = \frac{2p_1}{p_2}.
\end{aligned}
$$

Since aggregate endowment is $(2, 2)$,

$$
z(p) = \left(2\frac{p_2}{p_1} - 2,\ 2\frac{p_1}{p_2} - 2\right).
$$

Step 4: equilibrium. Set $z_1(p) = 0$:

$$
2\frac{p_2}{p_1} - 2 = 0 \quad \Longleftrightarrow \quad p_1 = p_2.
$$

Then $z_2(p) = 0$ automatically. Any positive multiple of $(1, 1)$ is a WE price. Under simplex normalization, $p^* = (1/2, 1/2)$.

Step 5: allocation. At $p_1 = p_2$, each agent has wealth 1 under simplex normalization or 2 if $p = (1, 1)$. In real terms the allocation is

$$
x_1^* = x_2^* = (1, 1).
$$

Exam-facing rule

On this ASU problem, the fastest safe route is: derive the generic square-root demand once, plug in $w_i = p \cdot \omega_i$, aggregate, then verify $p \cdot z(p) = 0$.

8.3 Walras' Law verification

For the square-root economy,

$$\begin{aligned} p \cdot z(p) &= p_1 \left(2\frac{p_2}{p_1} - 2\right) + p_2 \left(2\frac{p_1}{p_2} - 2\right) \\ &= 2p_2 - 2p_1 + 2p_1 - 2p_2 = 0. \end{aligned}$$

This is also a useful algebra check: if your excess demand does not satisfy Walras' Law, you likely treated wealth as fixed instead of $p \cdot \omega_i$.

9 Canonical Model II: Existence of WE Through Excess Demand

9.1 The DGKN existence checklist

Let $z : \Delta^\circ \to \mathbb{R}^L$ be a single-valued aggregate excess demand function. A standard sufficient condition for existence of a price $p^* \in \Delta^\circ$ with $z(p^*) = 0$ is:

1. **Continuity:** z is continuous on Δ°.
2. **Walras' Law:** $p \cdot z(p) = 0$ for all $p \in \Delta^\circ$.
3. **Boundedness below:** there exists $b \in \mathbb{R}^L$ such that $z(p) \geq b$ for all $p \in \Delta^\circ$.
4. **Boundary condition:** if $p^n \to p \in \Delta \setminus \Delta^\circ$, then $\|z(p^n)\| \to \infty$.

Mathematical intuition

Continuity prevents jumps. Walras' Law says aggregate budgets balance. Boundedness below says aggregate supply of any good cannot be infinitely negative. The boundary condition says that as a good's price goes to zero, some excess demand must explode, so the fixed-point construction cannot settle on the boundary.

Qualifier trap

The boundary condition is about *every sequence* approaching the boundary. A function can blow up near one boundary face but fail the boundary condition if it stays bounded along another boundary approach.

9.2 Proof skeleton via a price-adjustment correspondence

Define $\varphi : \Delta \rightrightarrows \Delta$ by

$$\varphi(p) = \begin{cases} \arg\max_{q \in \Delta} q \cdot z(p), & p \in \Delta^\circ, \\ \{q \in \Delta : p \cdot q = 0\}, & p \in \Delta \setminus \Delta^\circ. \end{cases}$$

For interior p, $\varphi(p)$ puts weight on goods with highest excess demand. For boundary p, it selects prices orthogonal to p, meaning it puts weight only on goods whose boundary price is zero.

Step 1: Kakutani. Under the four conditions, φ is nonempty-valued, convex-valued, closed-valued, and has a closed graph. Since Δ is nonempty, compact, and convex, Kakutani gives $p^* \in \Delta$ such that $p^* \in \varphi(p^*)$.

Step 2: fixed point cannot be on boundary. If $p^* \in \Delta \setminus \Delta^\circ$, then $p^* \in \varphi(p^*)$ implies $p^* \cdot p^* = 0$. Since $p^* \geq 0$, this implies $p^* = 0$, contradicting $p^* \in \Delta$. Hence $p^* \in \Delta^\circ$.

Step 3: fixed point clears markets. For any coordinate vector $e_\ell \in \Delta$,

$$e_\ell \cdot z(p^*) \leq p^* \cdot z(p^*) = 0,$$

where the inequality uses $p^* \in \arg\max_{q \in \Delta} q \cdot z(p^*)$ and the equality uses Walras' Law. Thus $z_\ell(p^*) \leq 0$ for every ℓ. If $z(p^*) < 0$ in at least one coordinate and never positive, then $p^* \cdot z(p^*) < 0$, contradicting Walras' Law. Therefore $z(p^*) = 0$.

9.3 ASU 2021 prototype: a function with no equilibrium

Let

$$z(p_1, p_2) = \left(-1, \frac{p_1}{p_2}\right), \qquad (p_1, p_2) \gg 0.$$

Continuity. Continuous on $\mathbb{R}^2_{++}$.

Homogeneity.

$$z(\lambda p_1, \lambda p_2) = \left(-1, \frac{\lambda p_1}{\lambda p_2}\right) = z(p_1, p_2)$$

for every $\lambda > 0$.

Walras' Law.

$$p \cdot z(p) = p_1(-1) + p_2 \frac{p_1}{p_2} = 0.$$

Bounded below. Since $z_1 = -1$ and $z_2 > 0$, take $b = (-1, 0)$.

Boundary condition. Normalize to Δ°, so $p_2 = 1 - p_1$. As $p_2 \to 0$, $z_2 = p_1/p_2 \to \infty$. But as $p_1 \to 0$ with $p_2 \to 1$,

$$z(p) \to (-1, 0),$$

so z remains bounded. The boundary condition fails.

No WE. There is no $p \gg 0$ with $z(p) = 0$ because $z_1(p) = -1$ for all p.

9.4 How to produce a no-WE example

An exam-safe no-WE answer must show more than "a theorem assumption fails." It must prove no price clears all markets. The template is:

1. Split into relative-price cases: $p_1 < p_2$, $p_1 = p_2$, $p_1 > p_2$.
2. Compute each consumer's demand correspondence in each case.
3. Show aggregate demand differs from aggregate endowment in every case.

Example: max/min economy. Let

$$u_1(x_1, x_2) = \max\{x_1, x_2\}, \quad \omega_1 = (1, 0),$$

$$u_2(x_1, x_2) = \min\{x_1, x_2\}, \quad \omega_2 = (0, 1).$$

Aggregate endowment is $(1, 1)$.

Consumer 2 demands equal quantities:

$$x_{21} = x_{22} = \frac{p_2}{p_1 + p_2}.$$

Consumer 1 has wealth p_1 and buys the good that gives the larger quantity per dollar.

- If $p_1 < p_2$, consumer 1 demands $(1, 0)$. Then aggregate demand for good 1 is $1+p_2/(p_1+p_2) > 1$.
- If $p_1 > p_2$, consumer 1 demands $(0, p_1/p_2)$. Then aggregate demand for good 1 is $p_2/(p_1+p_2) < 1$.
- If $p_1 = p_2$, consumer 1 demands either $(1, 0)$ or $(0, 1)$, while consumer 2 demands $(1/2, 1/2)$, so aggregate demand is either $(3/2, 1/2)$ or $(1/2, 3/2)$.

No positive price vector clears both markets.

10 Canonical Model III: Equilibrium Correspondences and Closed Graphs

A short proof question often defines a parameterized equilibrium correspondence and asks for closed graph.

Definition 9 (Closed graph)**.** *A correspondence $Q : A \rightrightarrows B$ has a closed graph if whenever $a^n \to a$, $b^n \to b$, and $b^n \in Q(a^n)$ for every n, then $b \in Q(a)$.*

Proposition 1 (Zero set of a continuous map has closed graph)**.** *Let $F : U \times W \to \mathbb{R}^m$ be continuous and define*

$$Q(w) = \{u \in U : F(u, w) = 0\}.$$

Then Q has a closed graph.

Proof. Take any sequence $w^n \to w$, $u^n \to u$, with $u^n \in Q(w^n)$. Then

$$F(u^n, w^n) = 0 \quad \text{for all } n.$$

By continuity,

$$F(u, w) = \lim_{n\to\infty} F(u^n, w^n) = 0.$$

Therefore $u \in Q(w)$. □

Exam-facing rule

Do not overcomplicate the 2018 closed-graph problem. Once $Q(\omega)$ is defined by $F(\hat{p}, \omega) = 0$, the proof is simply "limits preserve zeros under continuity."

11 Canonical Model IV: Pareto Optimality in an Edgeworth Box

11.1 Planner problem and KKT

For two consumers and two goods, a common method is to parametrize the Pareto set by utility promised to consumer 2:

$$\max_{x_1,x_2\geq 0} u_1(x_1) \quad \text{s.t.} \quad u_2(x_2)\geq \bar{u}_2, \qquad x_1+x_2\leq \bar{\omega}.$$

Under local nonsatiation, resource constraints bind at a PO, so $x_2=\bar{\omega}-x_1$. Then solve

$$\max_{0\leq x_1\leq \bar{\omega}} u_1(x_1) \quad \text{s.t.} \quad u_2(\bar{\omega}-x_1)\geq \bar{u}_2.$$

Interior smooth case. Let $x_1 \gg 0$ and $x_2 \gg 0$. The Lagrangian is

$$\mathcal{L}=u_1(x_1)+\lambda\left[u_2(\bar{\omega}-x_1)-\bar{u}_2\right].$$

The FOC for good ℓ is

$$\frac{\partial u_1(x_1)}{\partial x_\ell}-\lambda\frac{\partial u_2(x_2)}{\partial x_\ell}=0.$$

For goods 1 and 2,

$$\frac{u_{11}(x_1)}{u_{12}(x_1)}=\frac{u_{21}(x_2)}{u_{22}(x_2)}.$$

Boundary case. If $x_{i\ell}=0$ for some agent and good, the FOCs become KKT inequalities. A boundary PO can have $MRS_1\neq MRS_2$. In a diagram, the contract curve may run along the boundary of the Edgeworth box.

11.2 Worked Edgeworth example: Cobb-Douglas with identical endowments

Let both agents have endowment $(1/2,1/2)$, so $\bar{\omega}=(1,1)$. Let

$$u_1(x,y)=\log x+\log y, \qquad u_2(x,y)=\frac{1}{2}\log x+\log y.$$

At an interior PO, equate MRS. For consumer 1,

$$MRS_1=\frac{MU_x}{MU_y}=\frac{1/x_1}{1/y_1}=\frac{y_1}{x_1}.$$

For consumer 2, with $x_2=1-x_1$ and $y_2=1-y_1$,

$$MRS_2=\frac{(1/2)/x_2}{1/y_2}=\frac{y_2}{2x_2}.$$

The interior contract curve satisfies

$$\frac{y_1}{x_1}=\frac{1-y_1}{2(1-x_1)}.$$

Solve:

$$\begin{aligned}
2y_1(1-x_1)&=x_1(1-y_1),\\
2y_1-2x_1y_1&=x_1-x_1y_1,\\
y_1(2-x_1)&=x_1,\\
y_1&=\frac{x_1}{2-x_1}.
\end{aligned}$$

Together with boundary PO segments if relevant, this describes the Pareto set.

Walrasian equilibrium for the same example. Agent 1's Cobb-Douglas shares are $(1/2, 1/2)$. Agent 2's shares are $(1/3, 2/3)$ because the exponents are $(1/2, 1)$ and sum to $3/2$. Let $r = p_x/p_y$ and normalize $p_y = 1$. Each has wealth $w_i = (p_x + p_y)/2 = (r + 1)/2$.

$$x_1^1 = \frac{1}{2}\frac{w_1}{p_x} = \frac{r+1}{4r}, \qquad x_1^2 = \frac{1}{3}\frac{w_2}{p_x} = \frac{r+1}{6r}.$$

Market clearing for good x gives

$$\frac{r+1}{4r} + \frac{r+1}{6r} = 1.$$

Thus

$$\frac{5(r+1)}{12r} = 1 \quad \Longleftrightarrow \quad 5r + 5 = 12r \quad \Longleftrightarrow \quad r = \frac{5}{7}.$$

The WE relative price is $p_x/p_y = 5/7$.

11.3 Linear preferences: inequality method

Let

$$u_1(x, y) = x + y, \qquad u_2(x, y) = 2x + y, \qquad \bar{\omega} = (1, 1).$$

Consumer 2 values good x more than consumer 1, while both value good y equally. A PO cannot allocate positive good x to consumer 1 while consumer 2 holds positive good y: transfer a small x from 1 to 2 and a small y from 2 to 1. Consumer 1 is unchanged if $dy = dx$, consumer 2 gains because $2dx - dy = dx > 0$.

An efficient allocation must therefore put all good x with consumer 2 unless consumer 1 already has all good y and compensation is impossible at the boundary. The exam point is not the exact drawing but the method: use direct swaps and inequalities rather than invalid MRS equalities.

12 Canonical Model V: State-Contingent Edgeworth Boxes

12.1 States as commodities

With one physical good and two states, a consumption plan is $x_i = (x_{i1}, x_{i2}) \in \mathbb{R}^2_+$. The two state-contingent commodities are:

- good in state 1;
- good in state 2.

The Edgeworth box has width ω_1 and height ω_2, the aggregate endowment in each state.

If consumer i has subjective expected utility

$$U_i(x_i) = \pi_1 u_i(x_{i1}) + \pi_2 u_i(x_{i2}),$$

with common beliefs π and strictly concave u_i, the problem is formally a two-good exchange economy.

12.2 Interior risk-sharing PO

A weighted planner problem is

$$\max_{x_i \geq 0} \sum_i \alpha_i \left[\pi_1 u_i(x_{i1}) + \pi_2 u_i(x_{i2})\right] \quad \text{s.t.} \quad \sum_i x_{is} \leq \omega_s, \quad s = 1, 2.$$

At an interior PO, with resource multipliers $\nu_s > 0$,

$$\alpha_i \pi_s u_i'(x_{is}) = \nu_s, \qquad i = 1, \ldots, I, \quad s = 1, 2.$$

Hence for every i,

$$\frac{u_i'(x_{i1})}{u_i'(x_{i2})} = \frac{\nu_1}{\nu_2} \frac{\pi_2}{\pi_1}.$$

The right-hand side does not depend on i. Therefore all consumers have equalized state marginal-utility ratios.

12.3 No aggregate risk: $\omega_1 = \omega_2$

Suppose $I = 2$, common beliefs, strict concavity, and $\omega_1 = \omega_2 = \omega$. The ASU risk-sharing conclusion is that the Pareto set is

$$\left\{(x^1, x^2) : x^1 = (y, y),\ x^2 = (\omega - y, \omega - y),\ 0 \le y \le \omega\right\}.$$

Why. Aggregate resources are identical across states. If an allocation gives some consumer state-risk while aggregate risk is zero, then another consumer must hold the opposite risk. A mean-preserving smoothing transfer across states can weakly improve both and strictly improve at least one because u_i is strictly concave. Thus every PO gives each consumer constant consumption across states.

12.4 Aggregate state 1 higher: $\omega_1 > \omega_2$

At a positive interior PO with two consumers, $x_{i1} > x_{i2}$ for both $i = 1, 2$.

Proof. At an interior PO, the ratio

$$R_i = \frac{u_i'(x_{i1})}{u_i'(x_{i2})}$$

is the same across consumers up to the common factor $\nu_1\pi_2/(\nu_2\pi_1)$. Since u_i is strictly concave, u_i' is strictly decreasing. Hence:

$$x_{i1} > x_{i2} \iff R_i < 1, \qquad x_{i1} = x_{i2} \iff R_i = 1, \qquad x_{i1} < x_{i2} \iff R_i > 1.$$

Because $\omega_1 > \omega_2$, at least one consumer must have $x_{i1} > x_{i2}$. Then the common FOC ratio forces every positive consumer to have the same inequality direction, so every consumer has $x_{i1} > x_{i2}$. □

Qualifier trap

The positive-allocation qualifier matters. At a boundary, marginal utilities or KKT multipliers may not be comparable in the same way. Always check whether the problem says "positive PO" before using equality of FOC ratios.

13 First Welfare Theorem

13.1 Statement

Theorem 1 (First Welfare Theorem)**.** *Let (x^*, y^*, p^*) be a Walrasian equilibrium of a finite Arrow-Debreu economy. If preferences are complete, transitive, and locally nonsatiated, then (x^*, y^*) is Pareto optimal.*

Convexity is not required for FWT. Differentiability is not required. The proof is a price-inequality contradiction.

13.2 Reusable proof

Step 1: better bundles cost strictly more. Fix consumer i. If

$$x_i' \succ_i x_i^*,$$

then

$$p^* \cdot x_i' > p^* \cdot x_i^*.$$

Otherwise, if $p^* \cdot x_i' \leq p^* \cdot x_i^*$, then x_i' is affordable whenever x_i^* is affordable. This contradicts optimality of x_i^*.

Step 2: weakly better bundles cannot cost less. If

$$x_i' \succeq_i x_i^*,$$

then

$$p^* \cdot x_i' \geq p^* \cdot x_i^*.$$

Suppose instead $p^* \cdot x_i' < p^* \cdot x_i^*$. By local nonsatiation, for every small $\varepsilon > 0$ there is $\tilde{x}_i$ near x_i' with $\tilde{x}_i \succ_i x_i'$. By continuity of prices, for $\tilde{x}_i$ close enough to x_i', $p^* \cdot \tilde{x}_i < p^* \cdot x_i^*$. By transitivity, $\tilde{x}_i \succ_i x_i^*$, contradicting Step 1.

Step 3: contradiction. Suppose a feasible allocation (x', y') Pareto dominates (x^*, y^*). Then for every i,

$$x_i' \succeq_i x_i^*,$$

and for some k,

$$x_k' \succ_k x_k^*.$$

By Steps 1 and 2,

$$\sum_i p^* \cdot x_i' > \sum_i p^* \cdot x_i^*.$$

Using feasibility of (x^*, y^*),

$$\sum_i p^* \cdot x_i^* = p^* \cdot \sum_i \omega_i + \sum_j p^* \cdot y_j^*.$$

Using firm profit maximization,

$$p^* \cdot y_j^* \geq p^* \cdot y_j' \quad \text{for every } j.$$

Therefore

$$\begin{aligned}
\sum_i p^* \cdot x_i' &> p^* \cdot \sum_i \omega_i + \sum_j p^* \cdot y_j^* \\
&\geq p^* \cdot \sum_i \omega_i + \sum_j p^* \cdot y_j' \\
&= p^* \cdot \sum_i x_i',
\end{aligned}$$

where the last equality uses feasibility of (x', y'). This says

$$p^* \cdot \sum_i x'_i > p^* \cdot \sum_i x'_i,$$

a contradiction.

13.3 What assumptions do in the proof

Assumption	Role in FWT proof
Consumer optimality	Better bundles cannot be affordable.
Local nonsatiation	Weakly better and cheaper can be perturbed to strictly better and still cheaper.
Transitivity	If $\tilde{x}_i \succ_i x'_i$ and $x'_i \succeq_i x^*_i$, then $\tilde{x}_i \succ_i x^*_i$.
Firm profit maximization	Alternative production plans cannot have higher value at p^*.
Market clearing	Converts individual price inequalities into an aggregate feasibility contradiction.
Convexity	Not used. It belongs to SWT/existence, not FWT.

Qualifier trap

FWT does not say the equilibrium is fair, unique, interior, or maximizes total utility. It says no feasible allocation can make someone strictly better off without hurting someone else, under the model's complete-market feasible set.

14 Second Welfare Theorem

14.1 Statement and versions

The clean exam version has two steps.

Theorem 2 (Weak Second Welfare Theorem)**.** *Suppose production sets are convex and preferences are complete, transitive, convex, and locally nonsatiated. If (x^*, y^*) is Pareto optimal, then there exists $p^* \neq 0$ such that:*

1. *$p^* \cdot y^*_j \geq p^* \cdot y_j$ for all $y_j \in Y_j$ and all firms j;*
2. *if $x_i \succ_i x^*_i$, then $p^* \cdot x_i \geq p^* \cdot x^*_i$ for all consumers i;*
3. *(x^*, y^*) is feasible.*

This is a quasi-equilibrium support.

A stronger Second Welfare Theorem obtains a true WE with transfers if the support inequality can be strengthened so strictly preferred bundles are strictly more expensive, or if each consumer has some cheaper bundle and preferences are continuous/convex enough to rule out equality problems.

14.2 Separating-hyperplane proof

Define each consumer's strictly preferred set:

$$V_i = \{x_i \in X_i : x_i \succ_i x_i^*\}.$$

Define aggregate preferred consumption and aggregate production:

$$V = \sum_{i=1}^{I} V_i, \qquad Y = \sum_{j=1}^{J} Y_j.$$

Convex preferences imply each V_i is convex. Convex production sets imply Y is convex. Hence V and $\bar{\omega} + Y$ are convex.

No intersection. Because (x^*, y^*) is Pareto optimal,

$$V \cap (\bar{\omega} + Y) = \emptyset.$$

If not, there exists $\sum_i x_i \in V$ and $\sum_j y_j \in Y$ such that

$$\sum_i x_i = \bar{\omega} + \sum_j y_j.$$

Then every consumer is strictly better than at x_i^* and the allocation is feasible, contradicting weak PO. With strong PO, use weakly preferred sets plus one strict improvement; the idea is the same but notation is heavier.

Separate. By a separating hyperplane theorem, there exists $p^* \neq 0$ such that

$$p^* \cdot v \geq p^* \cdot (\bar{\omega} + y) \quad \text{for all } v \in V,\ y \in Y.$$

Using local nonsatiation, $\sum_i x_i^*$ lies in the closure of V. Since it is also equal to $\bar{\omega} + \sum_j y_j^*$, the separating inequalities imply

$$p^* \cdot z \geq p^* \cdot \sum_i x_i^* = p^* \cdot \left(\bar{\omega} + \sum_j y_j^*\right) \geq p^* \cdot (\bar{\omega} + y)$$

for all $z \in \overline{V}$ and $y \in Y$.

Firm support. Fix firm j and replace only its production plan:

$$y = y_j + \sum_{h \neq j} y_h^* \in Y.$$

Then

$$p^* \cdot \left(\bar{\omega} + y_j^* + \sum_{h \neq j} y_h^*\right) \geq p^* \cdot \left(\bar{\omega} + y_j + \sum_{h \neq j} y_h^*\right).$$

Cancel common terms to get

$$p^* \cdot y_j^* \geq p^* \cdot y_j.$$

Consumer support. Fix consumer i and let $x_i \succ_i x_i^*$. Then

$$x_i + \sum_{h \neq i} x_h^* \in V.$$

The separation inequality gives

$$p^* \cdot \left(x_i + \sum_{h \neq i} x_h^* \right) \geq p^* \cdot \left(x_i^* + \sum_{h \neq i} x_h^* \right),$$

so

$$p^* \cdot x_i \geq p^* \cdot x_i^*.$$

14.3 Transfers

Assume the support is strong enough that consumer i chooses x_i^* from her budget. Define transfers by

$$T_i^* = p^* \cdot x_i^* - p^* \cdot \omega_i - \sum_{j=1}^{J} \theta_{ij} p^* \cdot y_j^*.$$

Then x_i^* is exactly affordable. Transfers balance because

$$\begin{aligned}
\sum_i T_i^* &= p^* \cdot \sum_i x_i^* - p^* \cdot \sum_i \omega_i - \sum_i \sum_j \theta_{ij} p^* \cdot y_j^* \\
&= p^* \cdot \left(\sum_i \omega_i + \sum_j y_j^* \right) - p^* \cdot \sum_i \omega_i - \sum_j \left(\sum_i \theta_{ij} \right) p^* \cdot y_j^* \\
&= 0.
\end{aligned}$$

Thus the PO allocation is decentralized as a WE with transfers.

Exam-facing rule

For a transfer construction question, always write the transfer formula and then prove $\sum_i T_i = 0$. This is often the easiest way to earn points after identifying supporting prices.

14.4 Direct construction in a Cobb-Douglas Edgeworth box

Suppose an exam gives a target PO allocation x^* and asks you to support it with transfers without invoking SWT. Use this protocol:

1. Compute each consumer's MRS at x_i^*.
2. Set the price ratio equal to the common MRS.
3. Pick a normalization, such as $p_2 = 1$.
4. Set transfer $T_i = p \cdot x_i^* - p \cdot \omega_i$ in pure exchange.
5. Verify $\sum_i T_i = 0$ by feasibility.
6. Verify each x_i^* solves her utility maximization at the new wealth.

Example matching the 2015 style. Let

$$u_1(x_{11}, x_{12}) = (x_{11}x_{12})^{1/2}, \qquad u_2(x_{21}, x_{22}) = (x_{21}x_{22})^{1/2},$$

with $\omega_1 = (2, 1)$, $\omega_2 = (1, 2)$, and proposed allocation

$$x_1^* = (1, 1), \qquad x_2^* = (2, 2).$$

Both agents have $MRS = x_2/x_1$ for good 1 in terms of good 2. At x_1^*, $MRS_1 = 1$; at x_2^*, $MRS_2 = 1$. Take $p = (1, 1)$. Transfers are

$$T_1 = p \cdot x_1^* - p \cdot \omega_1 = 2 - 3 = -1,$$

$$T_2 = p \cdot x_2^* - p \cdot \omega_2 = 4 - 3 = 1.$$

They balance. With Cobb-Douglas equal shares, each agent demands half wealth in each good. Consumer 1's post-transfer wealth is 2, so demand is $(1, 1)$. Consumer 2's post-transfer wealth is 4, so demand is $(2, 2)$. The target is a WE with transfers.

Qualifier trap

In SWT, the point is not that the original endowments support the PO. Redistribution changes wealth, not preferences or prices. Do not force the supporting budget line to pass through the original endowment.

15 Canonical Model VI: Robinson Crusoe and Production Economies

15.1 Definition in one-consumer one-firm economy

A Robinson Crusoe economy has one consumer, one firm, endowment ω, production set Y, and ownership share 1. A competitive equilibrium is (x^*, y^*, p^*) such that:

1. $y^* \in \arg\max_{y \in Y} p^* \cdot y$;
2. $x^* \in \arg\max\{u(x) : x \in X,\ p^* \cdot x \leq p^* \cdot \omega + p^* \cdot y^*\}$;
3. $x^* = \omega + y^*$.

Qualifier trap

The consumer owns the firm. Profit income enters the consumer's budget even when there is only one consumer. In constant-returns examples, equilibrium often requires zero profit; otherwise the firm may have no finite profit maximizer.

15.2 Worked ASU-style Robinson Crusoe economy

The firm uses good 2 to produce good 1:

$$Y = \{(q, -z) : z \geq 0,\ Az \geq q\},$$

with $A > 0$. The consumer has utility

$$u(x_1, x_2) = 2\sqrt{x_1} + x_2,$$

endowment $(0, \omega)$, and owns the firm. Normalize the price of good 2 to 1 and write the price of good 1 as $p \geq 0$.

Firm problem. The firm chooses $z \geq 0$ and $q \leq Az$ to maximize

$$pq - z.$$

Given z, it chooses $q = Az$ if $p \geq 0$. Profit is

$$(pA - 1)z.$$

Thus the supply correspondence is:

$$Y^s(p) = \begin{cases} \{(0,0)\}, & pA < 1, \\ \{(Az, -z) : z \geq 0\}, & pA = 1, \\ \text{no finite maximizer}, & pA > 1. \end{cases}$$

Consumer demand. Let consumer wealth be $W = \omega + \pi(p)$, where $\pi(p)$ is firm profit. In equilibrium with $pA = 1$, profit is zero for all supplied z, so $W = \omega$. For a given $p > 0$, solve

$$\max_{x_1, x_2 \geq 0} 2\sqrt{x_1} + x_2 \quad \text{s.t.} \quad px_1 + x_2 \leq W.$$

If $x_2 > 0$, the FOC is

$$\frac{1}{\sqrt{x_1}} = p \quad \Longrightarrow \quad x_1 = \frac{1}{p^2}.$$

This costs $px_1 = 1/p$. Therefore

$$x_1^d(p, W) = \begin{cases} 1/p^2, & W \geq 1/p, \\ W/p, & W < 1/p, \end{cases}$$

and $x_2^d = W - px_1^d$.

Equilibrium. A finite equilibrium requires $pA = 1$, so $p^* = 1/A$ and $W = \omega$. Demand for good 1 is

$$x_1^d(p^*, \omega) = \begin{cases} A^2, & \omega \geq A, \\ A\omega, & \omega < A. \end{cases}$$

The firm must supply $q = x_1^d$ and use input $z = q/A$. Hence

$$z^* = \min\{A, \omega\}, \qquad q^* = A\min\{A, \omega\},$$

with consumer allocation

$$x^* = (q^*, \omega - z^*).$$

Verification. If $pA < 1$, the firm supplies no output, but the consumer demands positive good 1 for finite p, so good 1 does not clear. If $pA > 1$, the firm has no profit-maximizing production plan. Therefore all CE have $p^* = 1/A$ and the allocation above.

16 Core as Optional GE Extension

Definition 10 (Coalition and blocking)**.** *In a pure exchange economy with consumer set $\mathcal{J} = \{1, \ldots, I\}$, a coalition is any $S \subseteq \mathcal{J}$. Coalition S blocks feasible allocation x if there are bundles $(y_i)_{i \in S}$ such that*

$$\sum_{i \in S} y_i = \sum_{i \in S} \omega_i,$$

$y_i \succeq_i x_i$ for all $i \in S$, and $y_k \succ_k x_k$ for at least one $k \in S$.

Definition 11 (Core)**.** *The core is the set of feasible allocations not blocked by any coalition.*

Proposition 2. *Every core allocation is Pareto optimal.*

Proof. If a feasible allocation is not Pareto optimal, then the grand coalition $\mathcal{J}$ can reallocate the aggregate endowment to Pareto dominate it. Hence the grand coalition blocks it. Therefore a core allocation must be Pareto optimal. □

Proposition 3. *Under monotonicity, every Walrasian equilibrium allocation is in the core.*

Proof. Let (x^*, p^*) be a WE and suppose coalition S blocks it with $(y_i)_{i \in S}$. Since $y_i \succeq_i x_i^*$ for all $i \in S$ and $y_k \succ_k x_k^*$ for some k, consumer optimality implies

$$p^* \cdot y_i \geq p^* \cdot x_i^* \quad \text{for all } i \in S, \qquad p^* \cdot y_k > p^* \cdot x_k^*.$$

Summing,

$$p^* \cdot \sum_{i \in S} y_i > p^* \cdot \sum_{i \in S} x_i^*.$$

But blocking feasibility gives $\sum_{i \in S} y_i = \sum_{i \in S} \omega_i$, and each equilibrium allocation satisfies $p^* \cdot x_i^* = p^* \cdot \omega_i$. Contradiction. □

17 Exam Variations: How the Archetype Mutates

17.1 2020 Q3 and 2024 Q1: same computation, different grading emphasis

Both questions use the square-root two-agent exchange economy. The invariant part is demand and $z(p)$. The mutation is part (c)/(d): the exam asks not only for WE but also for existence conditions and no-WE examples. A complete answer therefore needs both algebra and theorem language.

17.2 2021 Q1: properties of a proposed excess demand

The exam gives z directly rather than preferences. The task is diagnostic: continuity, homogeneity, Walras' Law, boundedness below, boundary condition, and whether a zero exists. The high-pass answer distinguishes "fails boundary" from "there is no zero because $z_1 = -1$."

17.3 2018 Q3: closed graph of equilibrium prices

The exam normalizes $p_L = 1$ and defines $Q(\omega) = \{\hat{p} : F(\hat{p}, \omega) = 0\}$. Do not prove existence or uniqueness. Just state the sequential definition of closed graph and use continuity of F.

17.4 2015 Q1: PO and WE with transfers

The exam gives an Edgeworth-box economy, asks for a WE, asks to prove a proposed allocation is PO, then asks to support it with transfers. The invariant steps are: compute MRS, identify supporting price, construct transfers, verify demands.

17.5 2017 Q1: private ownership and missing market clearing

The exam says the first $L-1$ markets clear and asks to prove the Lth clears. This is Walras' Law in a private-ownership economy: include firm profits, transfers if present, and then use $p_L > 0$.

17.6 2022 Q1 and 2023 Q3: Robinson Crusoe

The variation is production. The invariant answer is firm profit maximization, consumer demand with profit income, and feasibility. The 2023 version has linear production and unbounded profit if price is wrong; the 2022 version asks about whether a WE necessarily exists and whether the SFWT assumptions hold.

17.7 2023 August Q1 and 2018 Q4: state-contingent boxes

The physical good is single, but states create commodities. Treat the problem as a two-good Edgeworth box. The common-belief strict-concavity structure gives risk-sharing restrictions that can be proved by FOCs or by smoothing arguments.

18 Common Traps and Fast Fixes

Trap	**Fast fix**
Forgetting endowment wealth.	Always write $w_i(p) = p \cdot \omega_i$ before solving demand.
Equating MRS at a boundary.	Start with KKT or direct improvement. MRS equality only for smooth interior PO.
Using convexity in FWT.	Replace with the price-inequality contradiction. Convexity belongs to SWT.
Claiming SWT gives WE without transfers.	It gives support after redistribution; write $T_i = p \cdot x_i^* - p \cdot \omega_i - \sum_j \theta_{ij} p \cdot y_j^*$.
Ignoring quasi-equilibrium distinction.	Weak support says strictly preferred bundles are not cheaper; true demand may require strictly more expensive.
Checking only one boundary sequence.	Boundary condition must hold for every sequence approaching any boundary face.
Forgetting price positivity in missing-market proof.	$p_k z_k = 0$ implies $z_k = 0$ only if $p_k > 0$.
Treating state-contingent PO as ex post PO.	Complete markets trade claims before the state; PO is ex ante with respect to preferences/beliefs.

Trap	Fast fix
Assuming no-WE follows from nonconvexity.	Nonconvexity may break existence theorem but does not prove nonexistence. Analyze demand cases.
Forgetting firm profit income.	In private ownership, consumer budget includes $\sum_j \theta_{ij} p \cdot y_j$.
Normalizing twice.	Use either $p_L = 1$ or $\sum p_\ell = 1$, not both unless requested.
Writing "market clears" as $z \leq 0$.	ASU WE definitions usually require equality unless free disposal is explicitly built into market clearing.

19 Exercise Bank

The distribution is approximately 30 percent mechanical, 30 percent proof/characterization, 25 percent past-qual applications, and 15 percent trap/economist-claim questions.

19.1 Mechanical drills

M1. Skill: write definitions. State a pure exchange economy, a feasible allocation, a Walrasian equilibrium, and a Pareto optimal allocation for I consumers and L goods.

M2. Skill: budget wealth. In a two-good exchange economy with $\omega_1 = (3, 1)$ and $\omega_2 = (1, 3)$, write consumer wealths at price $p = (p_1, p_2)$.

M3. Skill: Cobb-Douglas demand. If $u_i(x) = \alpha_i \log x_1 + (1 - \alpha_i) \log x_2$, derive $x_i(p, p \cdot \omega_i)$.

M4. Skill: excess demand. For two consumers with Cobb-Douglas shares $\alpha_1 = 1/2$, $\alpha_2 = 1/3$ and endowments $(1, 0), (0, 1)$, derive $z_1(p)$.

M5. Skill: Walras' Law. Show directly that your z in M4 satisfies $p \cdot z(p) = 0$.

M6. Skill: missing market. Suppose $L = 3$, $p \gg 0$, $p \cdot z(p) = 0$, and $z_1 = z_2 = 0$. Prove $z_3 = 0$.

M7. Skill: price normalization. Convert relative price $p_1/p_2 = 3$ into a simplex-normalized price.

M8. Skill: Edgeworth coordinates. Total endowment is $(5, 4)$ and consumer 1 receives $(2, 3)$. What is consumer 2's bundle in the Edgeworth box?

M9. Skill: supporting transfers. In pure exchange, given $p = (2, 1)$, $\omega_1 = (1, 1)$, $\omega_2 = (1, 1)$, and target $x_1^* = (1/2, 2)$, $x_2^* = (3/2, 0)$, compute transfers that make each target bundle exactly affordable.

19.2 Proof and characterization drills

P1. Skill: FWT price inequalities. Prove that if x_i^* solves the consumer problem and preferences are LNS, then $x_i' \succeq_i x_i^*$ implies $p \cdot x_i' \geq p \cdot x_i^*$.

P2. Skill: FWT. Prove the First Welfare Theorem in a pure exchange economy.

P3. Skill: SWT support. Define $V_i = \{x_i : x_i \succ_i x_i^*\}$ and prove $V \cap (\bar{\omega} + Y) = \emptyset$ at a PO.

P4. Skill: transfer balance. Prove $\sum_i T_i^* = 0$ for the SWT transfer formula.

P5. Skill: closed graph. Let $Q(a) = \{x : F(x,a) = 0\}$ with continuous F. Prove Q has closed graph.

P6. Skill: boundary condition. Explain why demand may remain bounded for a good whose price tends to zero without violating the boundary condition, provided another good's excess demand explodes.

P7. Skill: PO KKT. Write KKT conditions for a two-agent two-good planner problem with nonnegativity constraints.

P8. Skill: state-contingent risk sharing. In a two-state economy with common beliefs and strictly concave utilities, derive the interior PO marginal-utility ratio condition.

P9. Skill: core. Prove that every core allocation is Pareto optimal.

19.3 Past-qual-style applications

Q1. Skill: 2020/2024 excess demand. Re-derive the square-root economy $z(p)$ and identify all WE prices under $p_1 + p_2 = 1$.

Q2. Skill: 2021 excess-demand diagnostics. For $z(p_1, p_2) = (-1, p_1/p_2)$, check continuity, homogeneity, Walras' Law, boundedness below, boundary condition, and existence of zero.

Q3. Skill: 2018 closed graph. Let $F(\hat{p}, \omega)$ be continuous and $Q(\omega) = \{\hat{p} : F(\hat{p}, \omega) = 0\}$. Prove Q has closed graph.

Q4. Skill: 2015 transfers. With Cobb-Douglas utilities and target allocation $x_1^* = (1,1)$, $x_2^* = (2,2)$, endowments $\omega_1 = (2,1)$, $\omega_2 = (1,2)$, support the target with transfers.

Q5. Skill: 2023 Robinson. Solve the Robinson Crusoe economy with $q = Az$, utility $2\sqrt{x_1} + x_2$, and endowment $(0, \omega)$.

Q6. Skill: 2023 state PO. If $\omega_1 = \omega_2$ in a two-state risk-sharing economy, prove every PO gives each consumer constant consumption across states.

Q7. Skill: no-WE example. Analyze the max/min economy in Section 6.4 and prove no positive price vector clears markets.

Q8. Skill: abstract economy. Explain why a competitive equilibrium of an exchange economy is an equilibrium of the Arrow-Debreu auctioneer abstract economy when the auctioneer maximizes $p \cdot z$.

19.4 Trap and economist-claim questions

T1. Claim: "If an economy violates convexity, FWT fails." True or false? Explain.

T2. Claim: "If $L - 1$ markets clear, the last market clears." State the missing assumptions.

T3. Claim: "A PO allocation is always a WE for the original endowments." True or false? Explain using SWT.

T4. Claim: "If an excess demand function fails the boundary condition, no WE exists." True or false? Give the correct logical relation.

20 Answer Sketches

20.1 Mechanical drills

M1. Economy: $(L, \{X_i, \succeq_i, \omega_i\}_{i=1}^{I})$. Feasible: $x_i \in X_i$ and $\sum_i x_i = \sum_i \omega_i$. WE: $x_i^* \in D_i(p^*, p^* \cdot \omega_i)$ and markets clear. PO: no feasible allocation weakly improves all and strictly improves one.

M2. $w_1 = 3p_1 + p_2$ and $w_2 = p_1 + 3p_2$. Avoid treating wealth as 4 unless prices are both 1.

M3. Budget shares: $x_{i1} = \alpha_i(p \cdot \omega_i)/p_1$ and $x_{i2} = (1 - \alpha_i)(p \cdot \omega_i)/p_2$. This follows from FOCs and budget exhaustion.

M4. $w_1 = p_1$, $w_2 = p_2$. Thus $X_1 = (1/2)(p_1/p_1) + (1/3)(p_2/p_1) = 1/2 + p_2/(3p_1)$. Aggregate endowment of good 1 is 1, so $z_1 = -1/2 + p_2/(3p_1)$.

M5. Compute z_2 similarly or use budget shares; then verify $p_1 z_1 + p_2 z_2 = 0$. If not, recheck endowment wealth.

M6. $0 = p \cdot z = p_3 z_3$ because $z_1 = z_2 = 0$. Since $p_3 > 0$, $z_3 = 0$.

M7. $p_1 = 3p_2$ and $p_1 + p_2 = 1$ gives $p_2 = 1/4$, $p_1 = 3/4$.

M8. Consumer 2 receives $(5, 4) - (2, 3) = (3, 1)$.

M9. $T_1 = p \cdot x_1^* - p \cdot \omega_1 = (1 + 2) - 3 = 0$. $T_2 = p \cdot x_2^* - p \cdot \omega_2 = 3 - 3 = 0$. The target is already exactly affordable at this price.

20.2 Proof and characterization drills

P1. If $p \cdot x_i' < p \cdot x_i^*$, LNS gives a nearby $\tilde{x}_i \succ_i x_i'$ still cheaper than x_i^*. By transitivity $\tilde{x}_i \succ_i x_i^*$, contradicting optimality because $\tilde{x}_i$ is affordable.

P2. Suppose x' Pareto dominates x^*. Better/weakly better bundles imply $\sum_i p \cdot x_i' > \sum_i p \cdot x_i^*$. Feasibility gives both sums equal to $p \cdot \bar{\omega}$ in pure exchange. Contradiction.

P3. If $v \in V \cap (\bar{\omega} + Y)$, then $v = \sum_i x_i$ with each $x_i \succ_i x_i^*$ and also $v = \bar{\omega} + \sum_j y_j$ for some feasible production. This feasible allocation strictly improves all, contradicting PO.

P4. Sum $T_i^* = p \cdot x_i^* - p \cdot \omega_i - \sum_j \theta_{ij} p \cdot y_j^*$. Use feasibility and $\sum_i \theta_{ij} = 1$ to cancel terms.

P5. If $a^n \to a$, $x^n \to x$, and $x^n \in Q(a^n)$, then $F(x^n, a^n) = 0$. Continuity gives $F(x, a) = 0$, so $x \in Q(a)$.

P6. Boundary condition is about the norm of the whole vector $z(p^n)$. The demand for one zero-price good can stay bounded while another component of excess demand goes to infinity, so the norm still diverges.

P7. Use Lagrange multipliers for resource constraints and nonnegative multipliers for $x_{i\ell} \geq 0$. Interior goods satisfy equal weighted marginal utilities; boundary goods satisfy complementary slackness inequalities.

P8. Weighted planner FOCs: $\alpha_i \pi_s u_i'(x_{is}) = \nu_s$. Divide state 1 by state 2: $u_i'(x_{i1})/u_i'(x_{i2}) = (\nu_1/\nu_2)(\pi_2/\pi_1)$ for all i.

P9. If a core allocation were not PO, the grand coalition could implement a Pareto-dominating feasible allocation. That would block it, contradiction.

20.3 Past-qual-style applications

Q1. Generic demand is $x_1 = wp_2/[p_1(p_1+p_2)]$, $x_2 = wp_1/[p_2(p_1+p_2)]$. Plug $w_1 = 2p_1$, $w_2 = 2p_2$ to get $z = (2p_2/p_1 - 2, 2p_1/p_2 - 2)$. Simplex WE is $(1/2, 1/2)$.

Q2. It is continuous, homogeneous degree zero, satisfies Walras' Law, and bounded below by $(-1, 0)$. It fails the boundary condition as $p_1 \to 0$. No zero exists because the first coordinate is always -1.

Q3. Use the zero-set proof: $F(\hat{p}^n, \omega^n) = 0$, convergence plus continuity gives $F(\hat{p}, \omega) = 0$.

Q4. At $(1,1)$ and $(2,2)$ MRSs equal 1, so take $p = (1,1)$. Transfers: $T_1 = -1$, $T_2 = 1$. Cobb-Douglas equal-share demands at wealths 2 and 4 are the target bundles.

Q5. Firm requires $p = 1/A$. Consumer demand at this price is A^2 if $\omega \geq A$, otherwise $A\omega$. Choose firm input $z^* = \min\{A, \omega\}$ and output $q^* = Az^*$.

Q6. With no aggregate risk, any individual risk must be offset by opposite risk for another consumer. Strict concavity lets the planner smooth both across states, a Pareto improvement. Thus PO allocations give constant state consumption to each consumer.

Q7. Split cases $p_1 < p_2$, $p_1 = p_2$, $p_1 > p_2$. Consumer 1 buys the cheaper max good; consumer 2 buys equal Leontief amounts. Aggregate demand differs from $(1,1)$ in every case.

Q8. At CE, consumers maximize and markets clear, so $p^* \cdot z^* = 0$. Since $p^* \in \Delta$ maximizes $p \cdot z^*$ over Δ and all components of z^* are zero, the auctioneer is optimizing too.

20.4 Trap questions

T1. False. FWT does not require convexity. Convexity is central to SWT and standard existence theorems.

T2. Need Walras' Law, the missing market's price strictly positive, and the first $L-1$ excess demands equal zero. In private ownership, Walras' Law must include profits and transfers.

T3. False. SWT says a PO can be supported after suitable transfers under convexity and other conditions. It need not be a WE for original endowments.

T4. False. The boundary condition is sufficient as part of an existence theorem, not necessary. Failure of a sufficient condition does not imply nonexistence.

21 Past-Qual Practice Map

Exam	Problem	How to use it
August 2024	Q1	Primary excess-demand computation; existence conditions; no-WE example.
June 2020	Q3	Same square-root exchange computation; good timed drill before 2024 Q1.

Exam	Problem	How to use it
June 2020	Q4	Definitions of exchange economy and CE; auctioneer abstract economy; relation between CE and abstract equilibrium.
June 2021	Q1	Diagnostic drill for excess-demand properties and failure of boundary condition.
August 2018	Q3	Closed-graph proof for equilibrium price correspondence.
August 2018	Q4	State-contingent Edgeworth box, no aggregate risk, equilibrium with $p = \pi$.
August 2015	Q1	Edgeworth WE, PO proof, and direct WE-with-transfers construction.
August 2017	Q1	Walras' Law: first $L - 1$ markets clear implies last market clears in private ownership economy.
June 2022	Q1	Robinson Crusoe with $Y = \mathbb{R}^L_-$; existence and SFWT assumption checks.
June 2023	Q3	Robinson Crusoe supply and demand; representative-firm discussion.
August 2023	Q1	State-contingent PO and risk-sharing proof; positive versus boundary cases.
June 2016	Q1	Missing/nontraded market and constrained PO versus full-market PO.
Problem Set 1	Exercises 1–3	Nonstandard boxes; indivisibilities; private ownership graphing.
Problem Set 2	Exercises 2–5	3-good WE; planner PO; separating hyperplane; Edgeworth computations.
Problem Set 3	Exercises 1–4	Direct SWT/existence; correspondences; boundary condition; failure examples.
Problem Set 4	Exercises 1–4	Boundary-condition proof; Gorman aggregation; core; demand proof.
Problem Set 5	Exercises 1,3	Equilibrium correspondence UHC; identical-consumer WE examples.

22 Final Study Checklist

Before attempting old GE quals, verify that you can do each item without notes.

1. State WE, WE with transfers, PO, quasi-equilibrium, aggregate excess demand, and Walras' Law.
2. Solve the square-root exchange economy and get $z = (2p_2/p_1 - 2, 2p_1/p_2 - 2)$.
3. State the DGKN existence conditions and explain the boundary condition in words.

4. Give a no-WE example and prove no price clears markets by cases.
5. Prove missing market clearing from Walras' Law.
6. Define closed graph and prove the zero-set closed-graph result.
7. Draw an Edgeworth box, label both origins, an endowment, a budget line, a PO point, and a transfer-supported budget line.
8. Write the planner problem for PO and derive $MRS_1 = MRS_2$ only under smooth interior conditions.
9. Prove FWT without using convexity.
10. Prove the SWT support theorem using preferred sets and separation.
11. Construct transfers and prove they balance.
12. Treat states as commodities and derive the risk-sharing FOC.
13. Solve the Robinson Crusoe linear-technology example and explain why wrong prices fail.
14. Identify when boundary, kink, nonconvexity, indivisibility, missing markets, or zero prices invalidate the standard shortcut.

23 One-Page Formula Sheet

High-yield formulas

$$B_i(p, w_i) = \{x_i \in X_i : p \cdot x_i \le w_i\}, \qquad w_i = p \cdot \omega_i + T_i + \sum_j \theta_{ij} p \cdot y_j,$$

$$z(p) = \sum_i \big(x_i(p, p \cdot \omega_i) - \omega_i\big),$$

$$p \cdot z(p) = 0 \quad \text{(Walras' Law)},$$

$$\Delta = \{p \in \mathbb{R}^L_+ : \sum_\ell p_\ell = 1\}, \qquad \Delta^\circ = \{p \in \Delta : p \gg 0\},$$

$$\text{DGKN existence: continuity + Walras' Law + bounded below + boundary condition},$$

$$\text{Interior PO: } MRS_i = MRS_j,$$

$$\text{Risk-sharing PO: } \frac{u_i'(x_{i1})}{u_i'(x_{i2})} = \frac{\nu_1}{\nu_2}\frac{\pi_2}{\pi_1},$$

$$T_i^* = p^* \cdot x_i^* - p^* \cdot \omega_i - \sum_j \theta_{ij} p^* \cdot y_j^*,$$

$$\text{Square-root demand: } x_1 = \frac{wp_2}{p_1(p_1 + p_2)}, \qquad x_2 = \frac{wp_1}{p_2(p_1 + p_2)}.$$